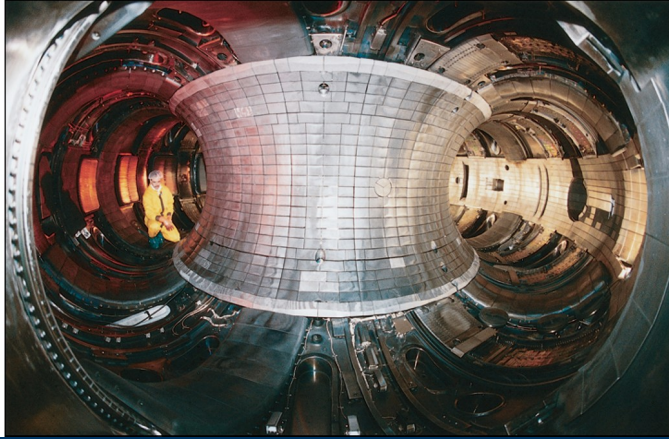




Topic 19

Magnetism

Reading Question 19-1 (1 of 2)

A particle of charge q travels with velocity v in the same direction as the magnetic field, B . What is the force exerted on the charge?

1. qvB , in the direction of the magnetic field
2. qvB , in the direction opposite that of the magnetic field
3. qvB , in a direction perpendicular to the magnetic field and the velocity
4. zero

Reading Question 19-1 (2 of 2)

A particle of charge q travels with velocity v in the same direction as the magnetic field, B . What is the force exerted on the charge?

1. qvB , in the direction of the magnetic field
2. qvB , in the direction opposite that of the magnetic field
3. qvB , in a direction perpendicular to the magnetic field and the velocity
4. **Answer: zero**

Reading Question 19-2 (1 of 2)

A particle of positive charge q travels with velocity v towards the top of the screen. The magnetic field present, B , points rightward. The force

1. points leftward.
2. points into the screen.
3. points toward the bottom of the screen.
4. points out of the screen.

Reading Question 19-2 (2 of 2)

A particle of positive charge q travels with velocity v towards the top of the screen. The magnetic field present, B , points rightward. The force

1. points leftward.
- 2. Answer: points into the screen.**
3. points toward the bottom of the screen.
4. points out of the screen.

Reading Question 19-3 (1 of 2)

Two parallel wires carry currents in the same direction and exert a force on each of magnitude F for each meter of length. One of the currents is now reversed in direction and doubled in magnitude. What are the resulting magnitude for the force per meter of length and its direction relative to the wires?

1. $2F$, attractive
2. $2F$, repulsive
3. $4F$, attractive
4. $4F$, repulsive

Reading Question 19-3 (2 of 2)

Two parallel wires carry currents in the same direction and exert a force on each of magnitude F for each meter of length. One of the currents is now reversed in direction and doubled in magnitude. What are the resulting magnitude for the force per meter of length and its direction relative to the wires?

1. $2F$, attractive
- 2. Answer: $2F$, repulsive**
3. $4F$, attractive
4. $4F$, repulsive

Reading Question 19-4 (1 of 2)

A charged particle is moving perpendicular to a uniform magnetic field. The radius of its path can be decreased most by

1. decreasing the magnitude of the magnetic field.
2. increasing its velocity.
3. decreasing its velocity and increasing the magnitude of the magnetic field.
4. increasing its velocity and decreasing the magnitude of the magnetic field.

Reading Question 19-4 (2 of 2)

A charged particle is moving perpendicular to a uniform magnetic field. The radius of its path can be decreased most by

1. decreasing the magnitude of the magnetic field.
2. increasing its velocity.
- 3. Answer: decreasing its velocity and increasing the magnitude of the magnetic field.**
4. increasing its velocity and decreasing the magnitude of the magnetic field.

Reading Question 19-5 (1 of 2)

Suppose a long, straight wire carries a current upwards. A short distance directly east of the wire, a negatively charge particle is moving upwards parallel to the wire. In what direction is the magnetic force on the particle?

1. The force is upwards.
2. The force is eastward.
3. The force is westward.
4. Either there is no force or its direction is not given among these choices.

Reading Question 19-5 (2 of 2)

Suppose a long, straight wire carries a current upwards. A short distance directly east of the wire, a negatively charge particle is moving upwards parallel to the wire. In what direction is the magnetic force on the particle?

1. The force is upwards.
- 2. Answer: The force is eastward.**
3. The force is westward.
4. Either there is no force or its direction is not given among these choices.

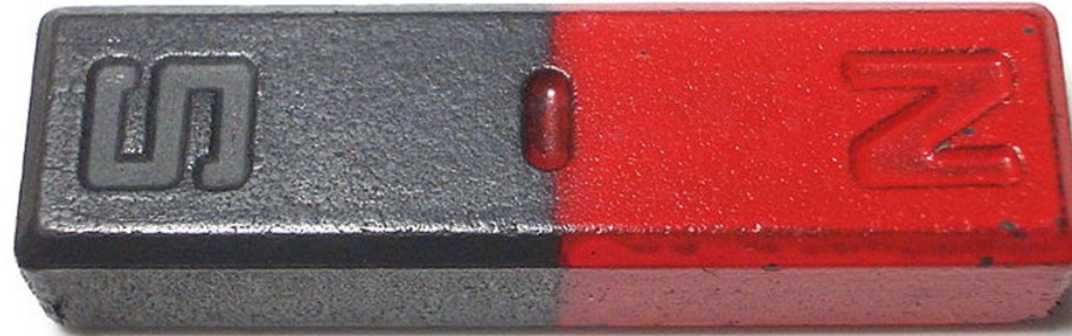
Topic 19: Magnetism

TOPIC DISCUSSION

Magnets (1 of 5)



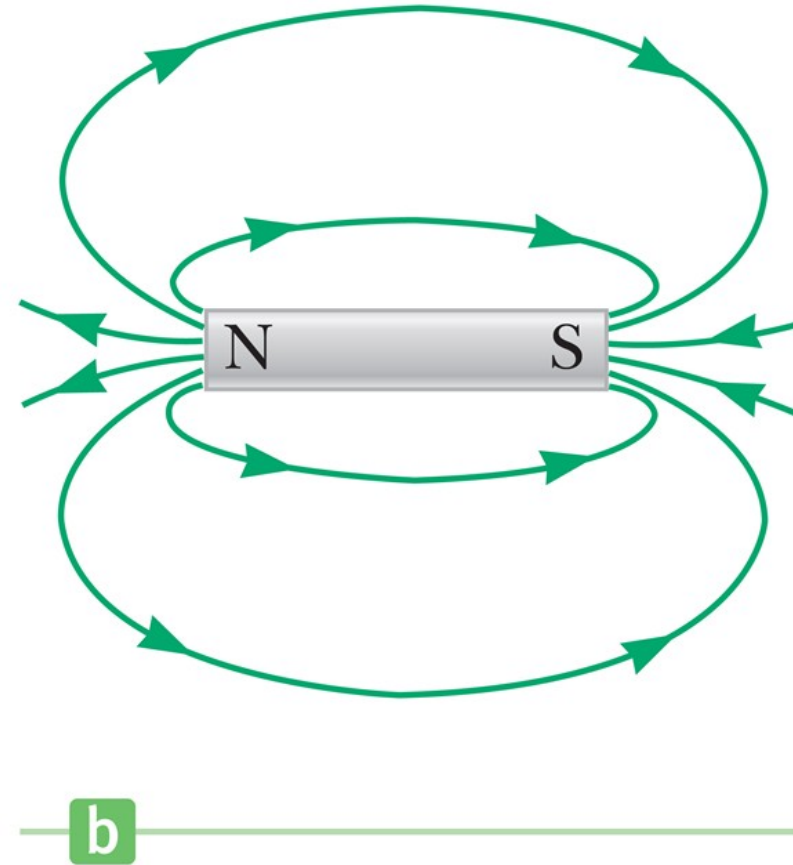
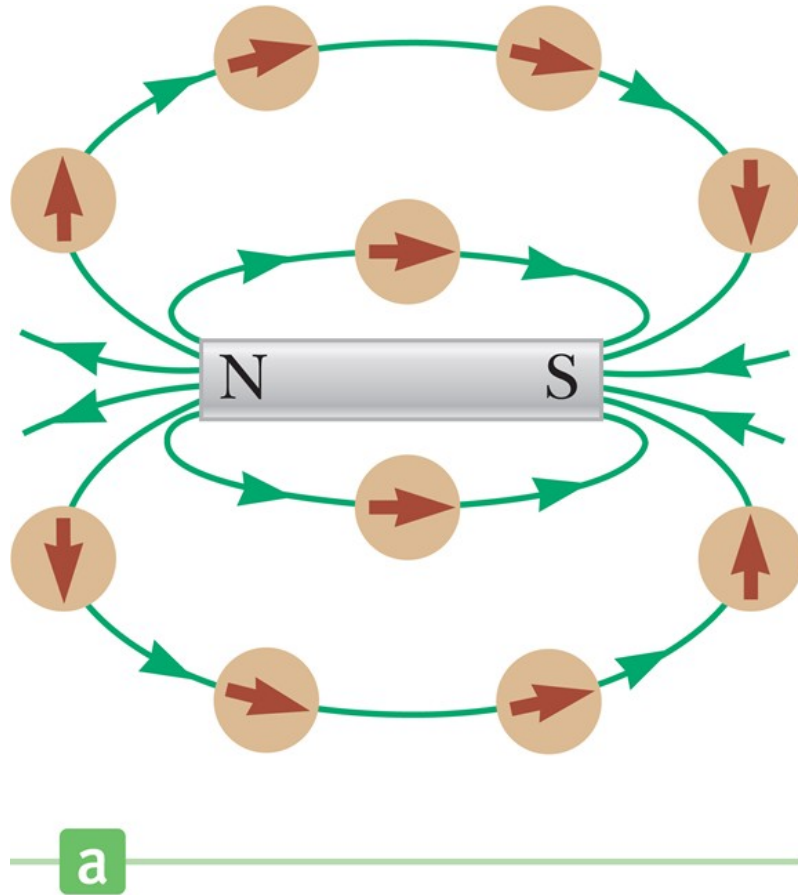
Magnets (2 of 5)



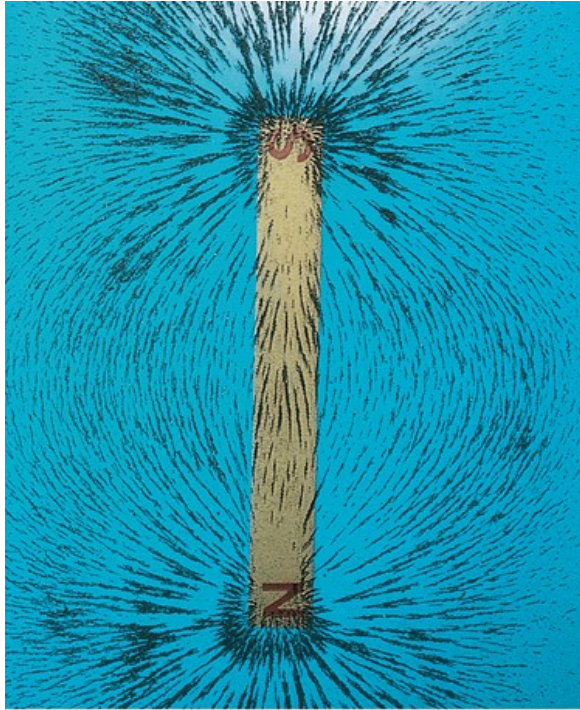
Magnets (3 of 5)



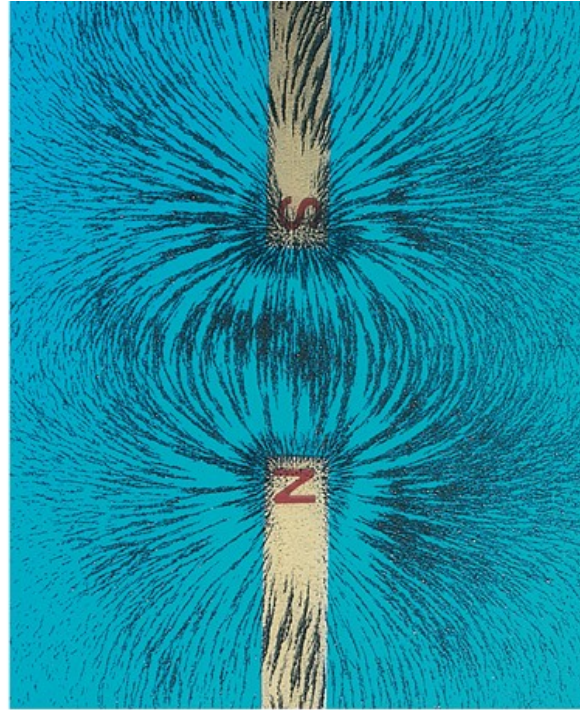
Magnets (4 of 5)



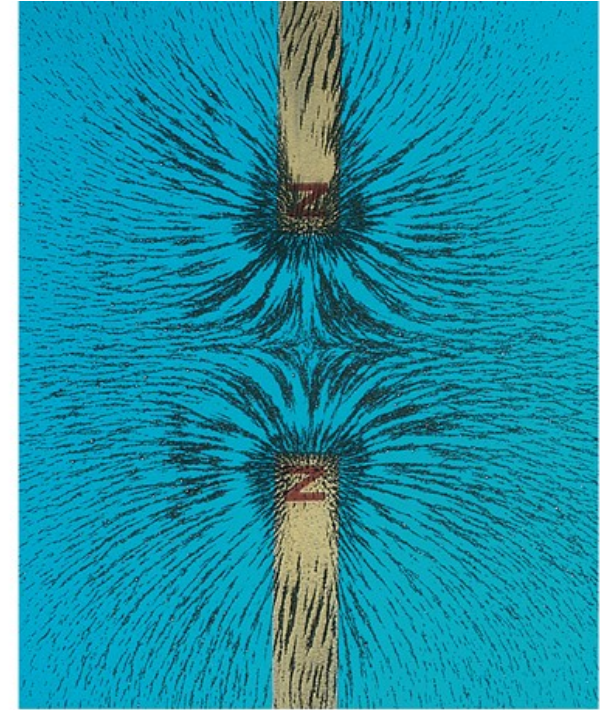
Magnets (5 of 5)



a



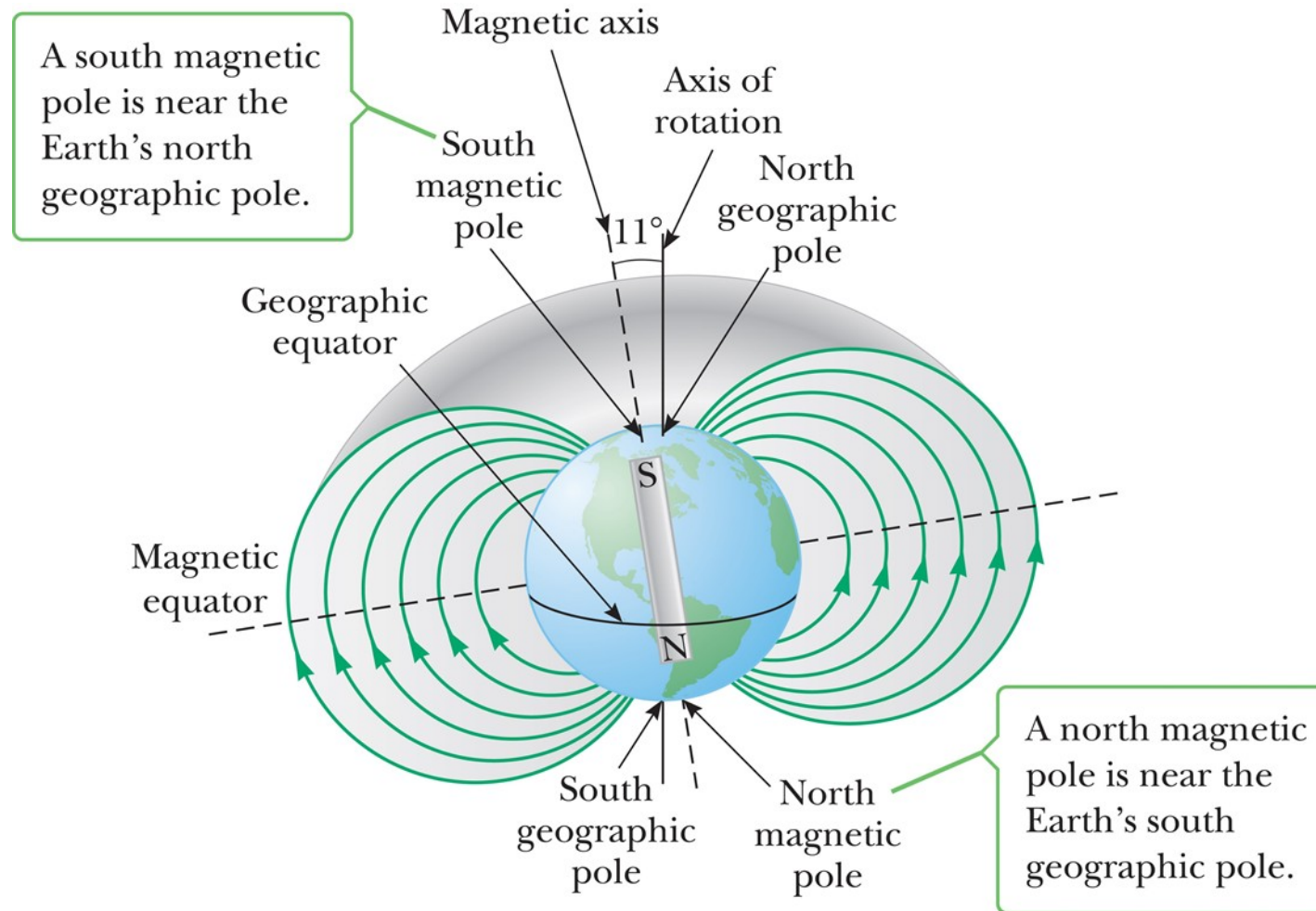
b



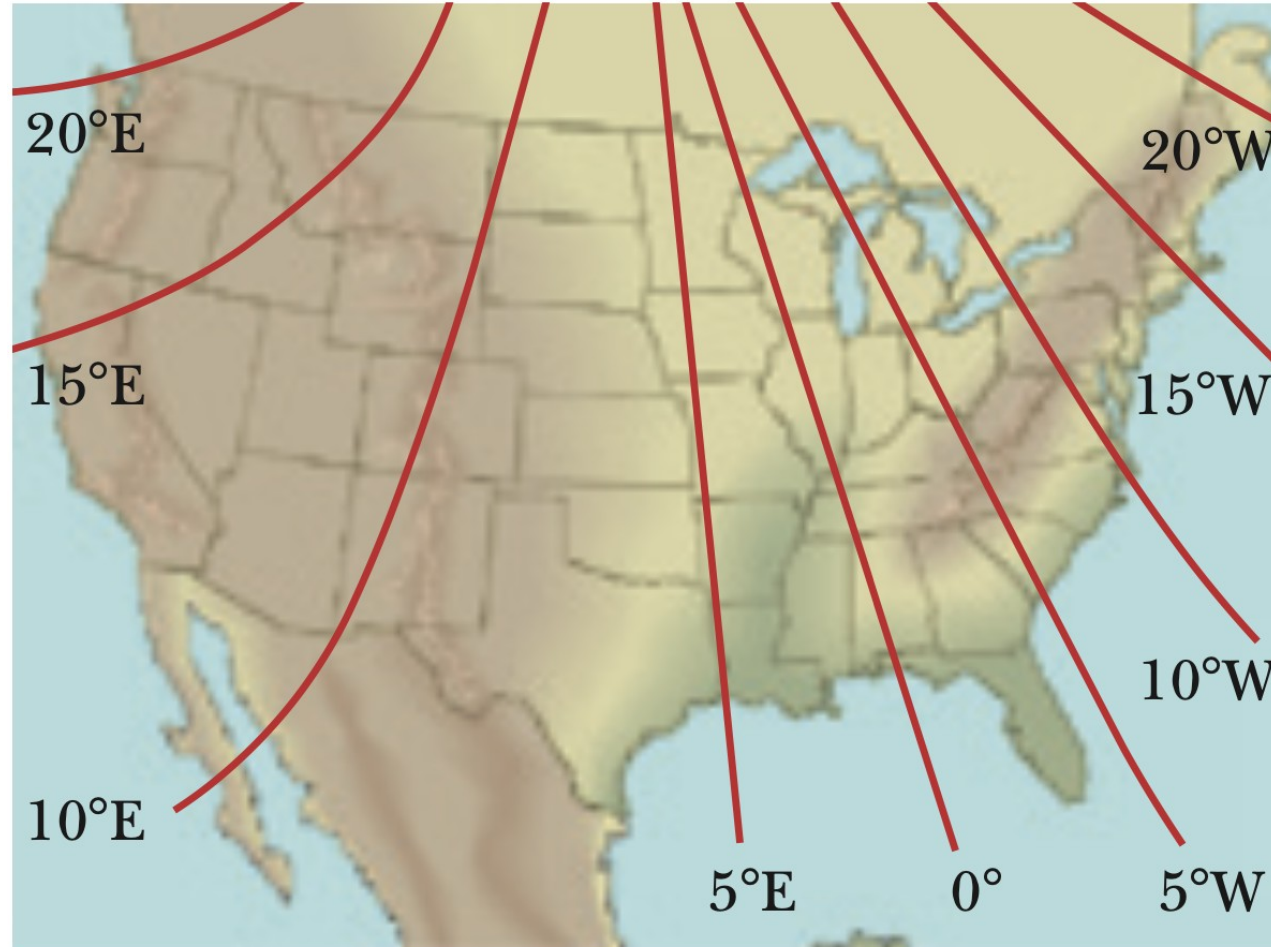
c

Courtesy of Henry Leap and Jim Lehman

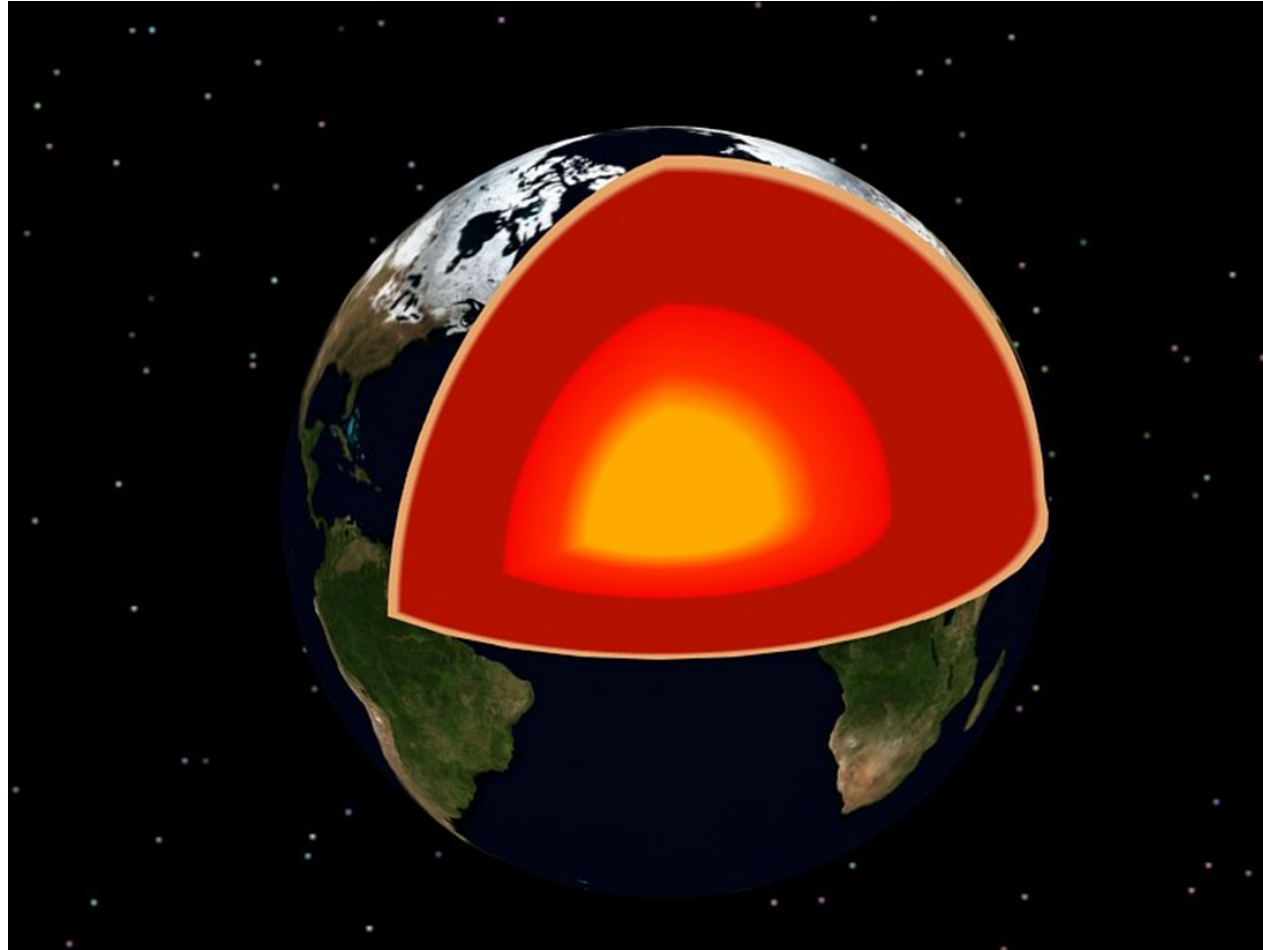
Earth's Magnetic Field (1 of 3)



Earth's Magnetic Field (2 of 3)



Earth's Magnetic Field (3 of 3)



Magnetic Bacteria



Labeling Airport Runways



Magnetic Fields (1 of 2)

$$F = qvB \sin \theta$$

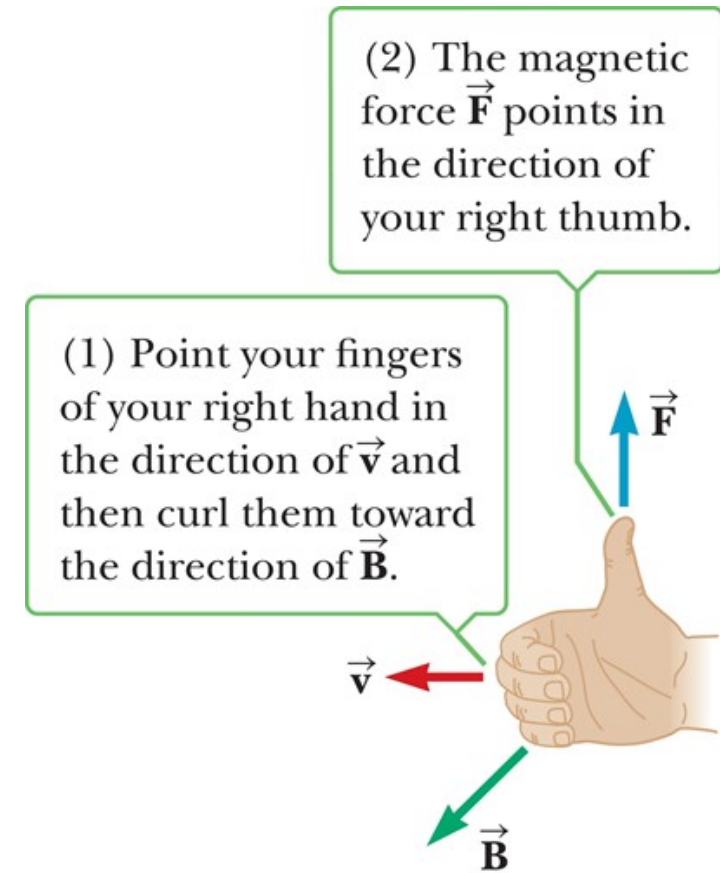
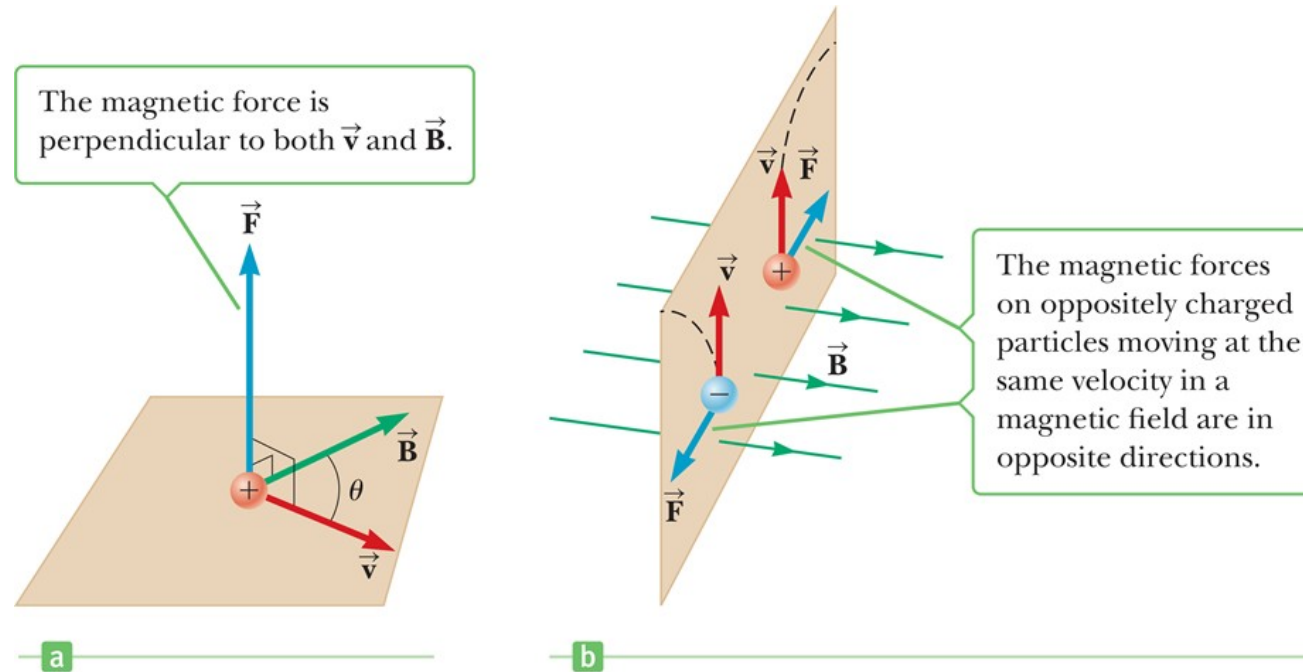
$$B \equiv \frac{F}{qv \sin \theta} \quad \text{SI unit: T or Wb/m}^2$$

$$[B] = \text{T} = \frac{\text{Wb}}{\text{m}^2} = \frac{\text{N}}{\text{C} \cdot \text{m/s}^2} = \frac{\text{N}}{\text{A} \cdot \text{m}}$$

$$1 \text{ T} = 10^4 \text{ G}$$

$$F_{\text{max}} = qvB$$

Magnetic Fields (2 of 2)



Think – Pair – Share 1

A charged particle moves in a straight line through a region of space. Which of the following answers *must* be true? (Assume any other fields are negligible.)

The magnetic field

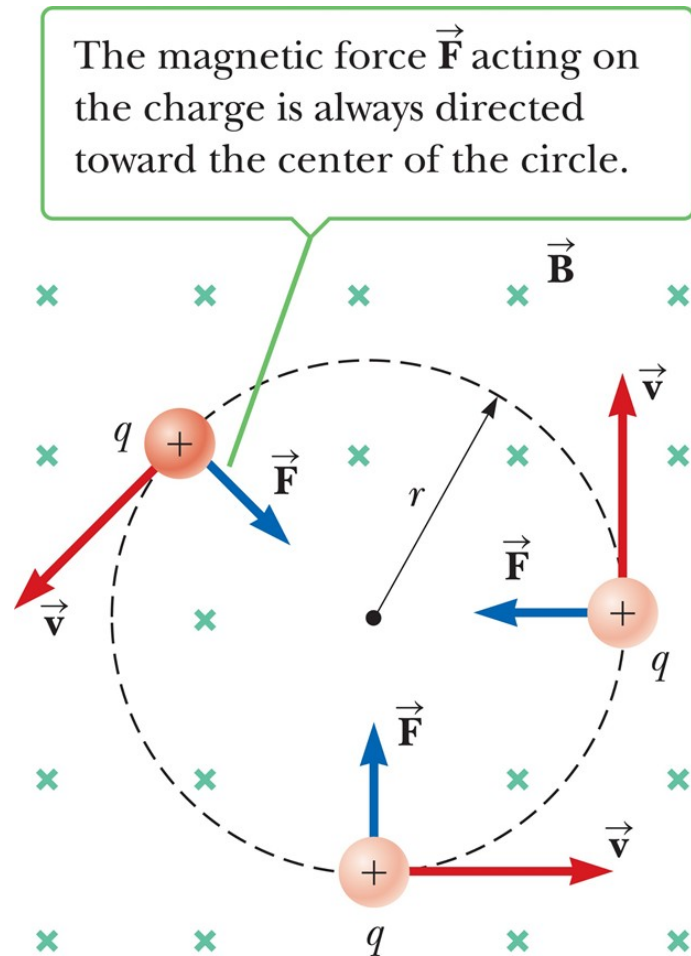
1. has a magnitude of zero.
2. has a zero component perpendicular to the particle's velocity.
3. has a zero component parallel to the particle's velocity.

Think – Pair – Share 2

The north-pole end of a bar magnet is held near a stationary positively charged piece of plastic. The plastic is

1. attracted.
2. repelled.
3. unaffected by the magnet.

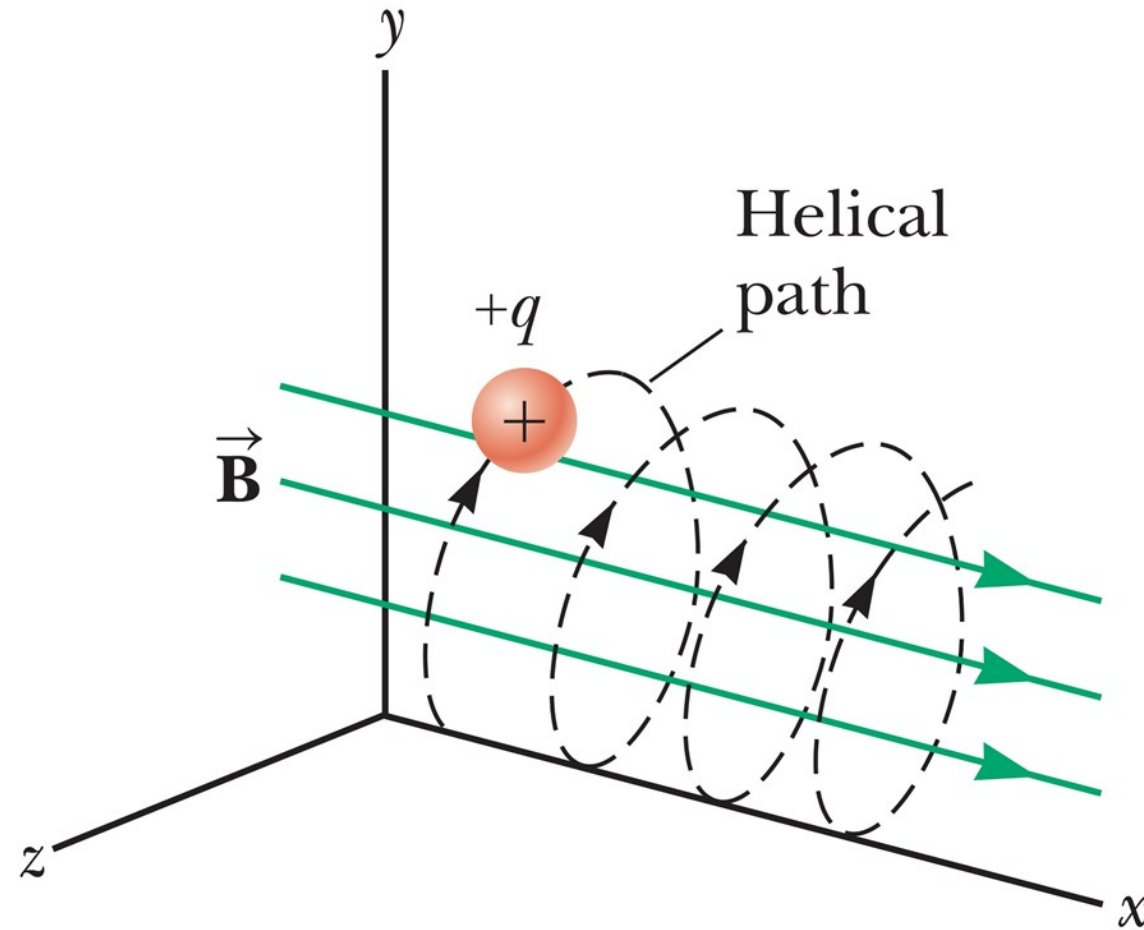
Motion of a Charged Particle in a Magnetic Field (1 of 2)



$$F = qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

Motion of a Charged Particle in a Magnetic Field (2 of 2)

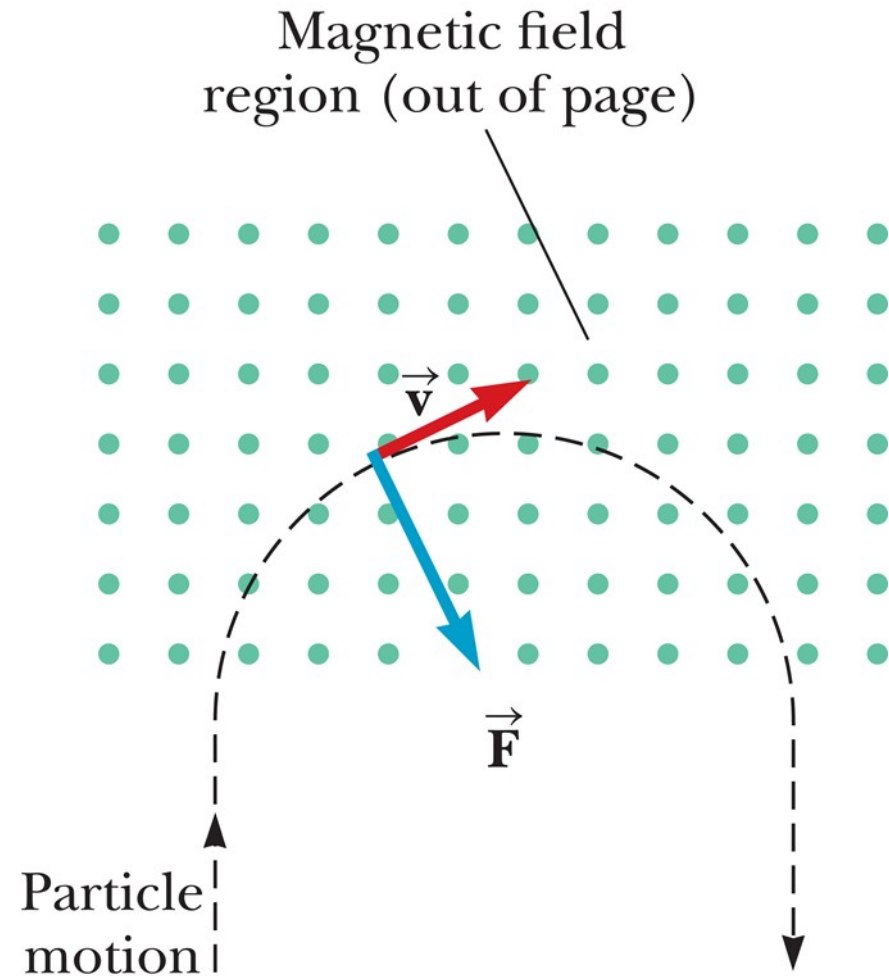
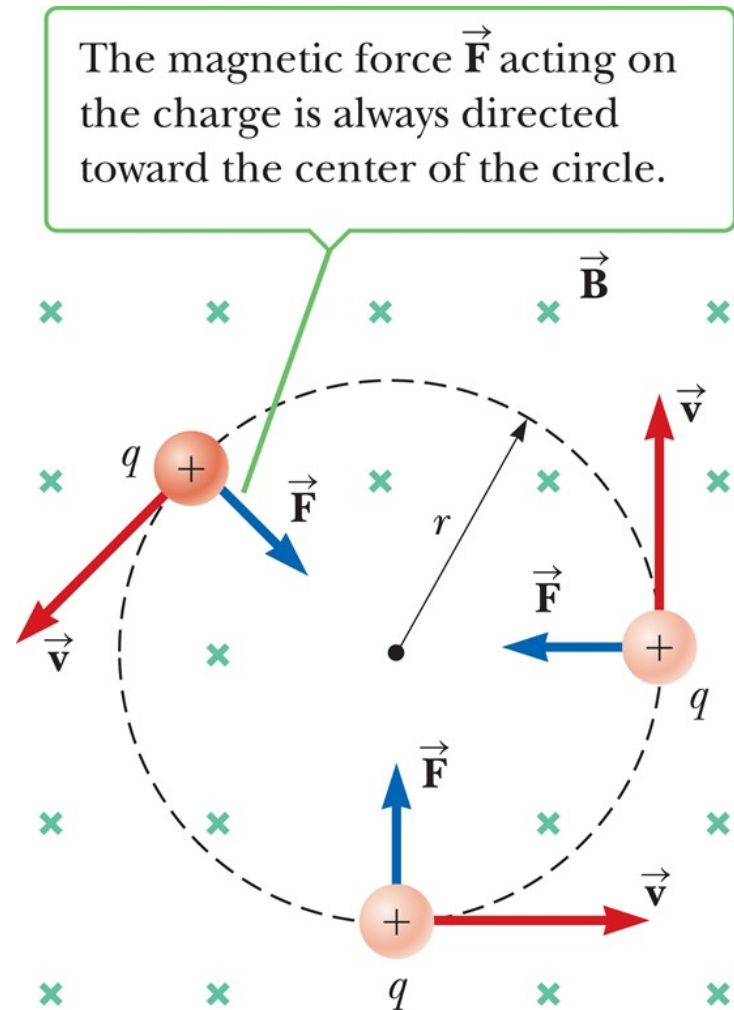


Think – Pair – Share 3

As a charged particle moves freely in a circular path in the presence of a constant magnetic field applied perpendicular to the particle's velocity, the particle's kinetic energy

1. remains constant.
2. increases.
3. decreases.

Notation

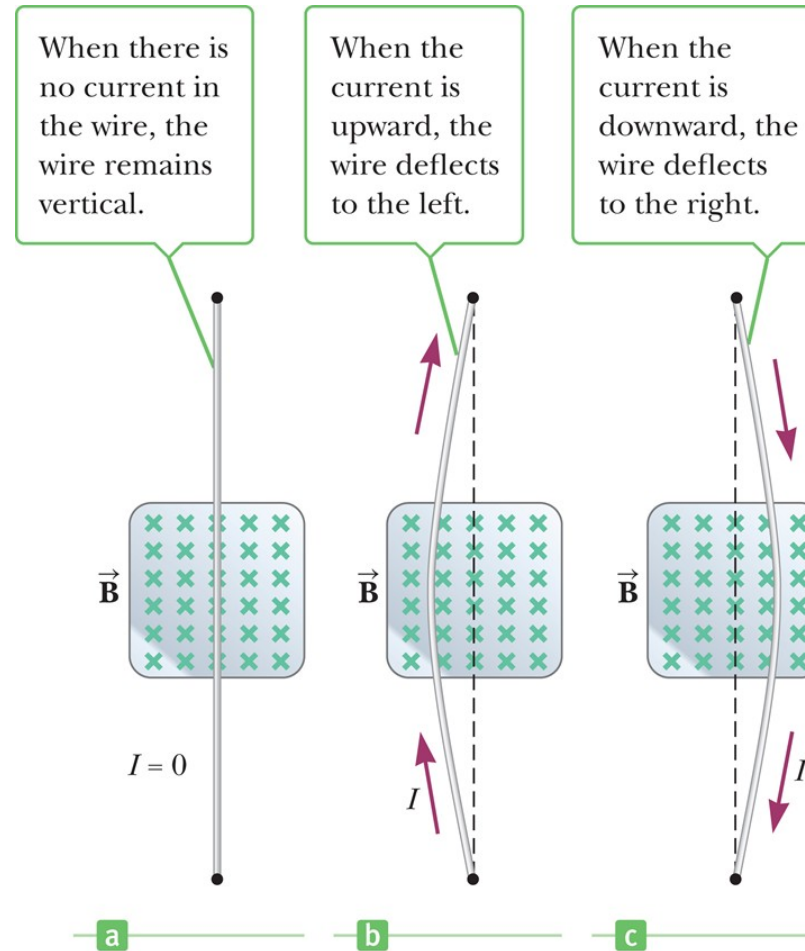


Magnetic Force on a Current-Carrying Conductor (1 of 6)

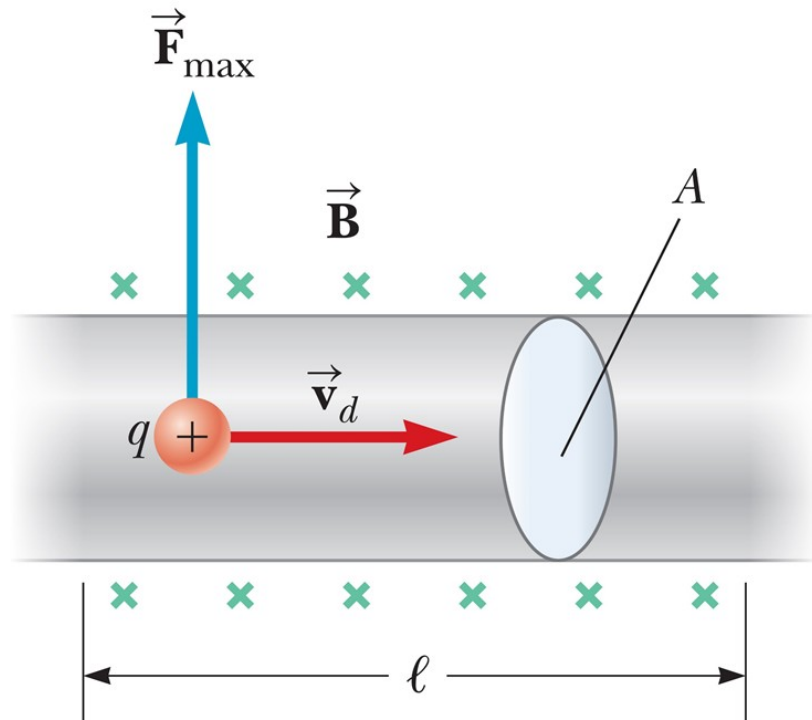
Courtesy of Henry Leap and Jim Lehman



Magnetic Force on a Current-Carrying Conductor (2 of 6)



Magnetic Force on a Current-Carrying Conductor (3 of 6)



$$F = qv_d B$$

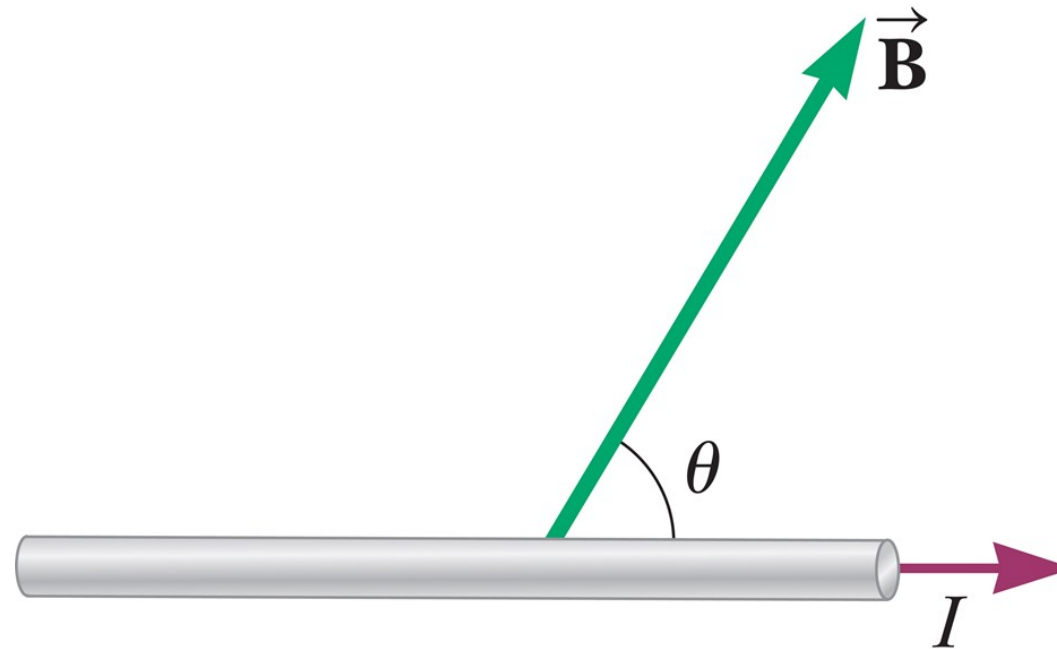
$$\text{volume} = A\ell$$

$$\text{number of carriers} = nA\ell$$

$$F_{\text{max}} = (qv_d B)(nA\ell)$$

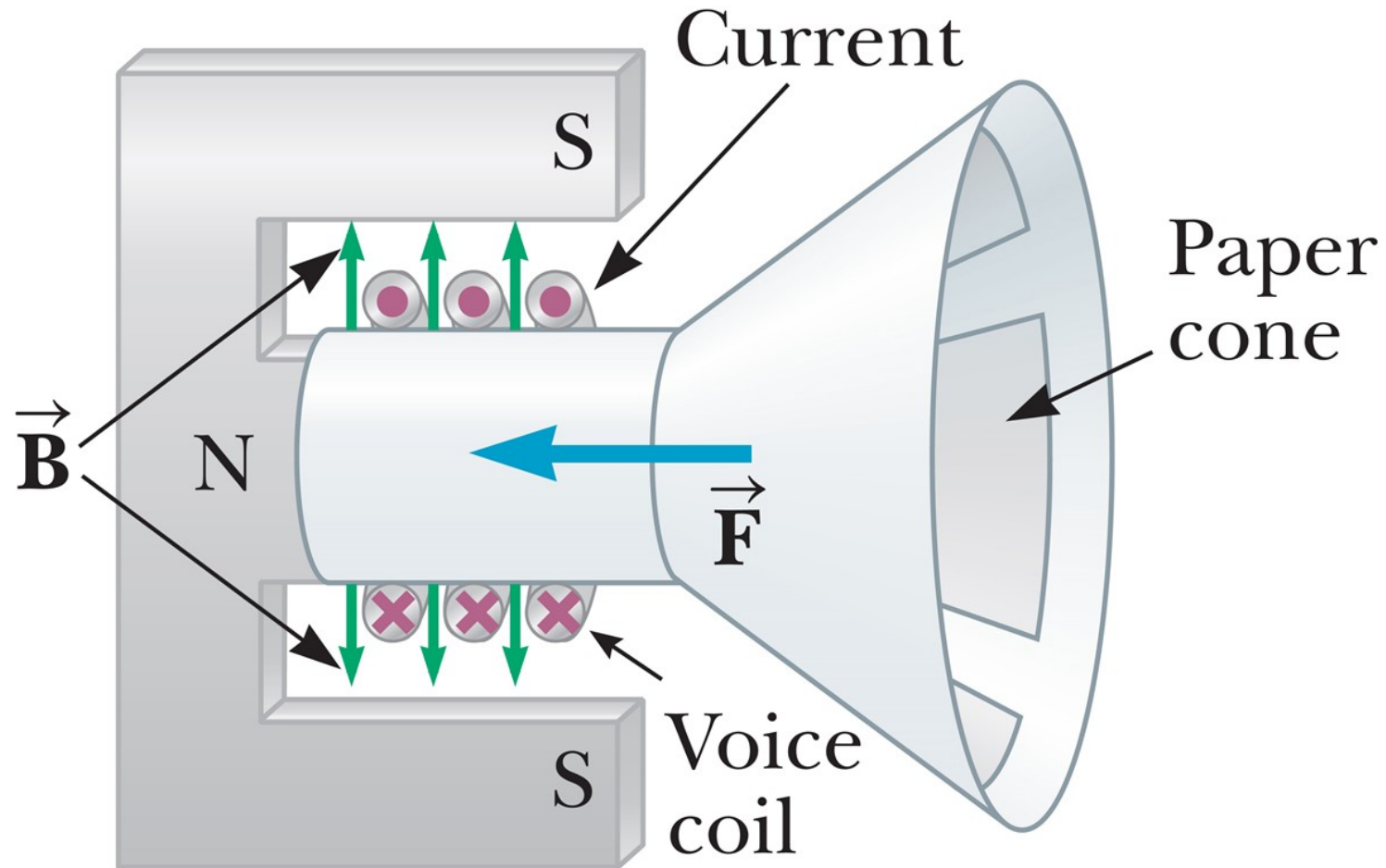
Magnetic Force on a Current-Carrying Conductor (4 of 6)

$$I = n q v_d A \rightarrow B I \ell$$

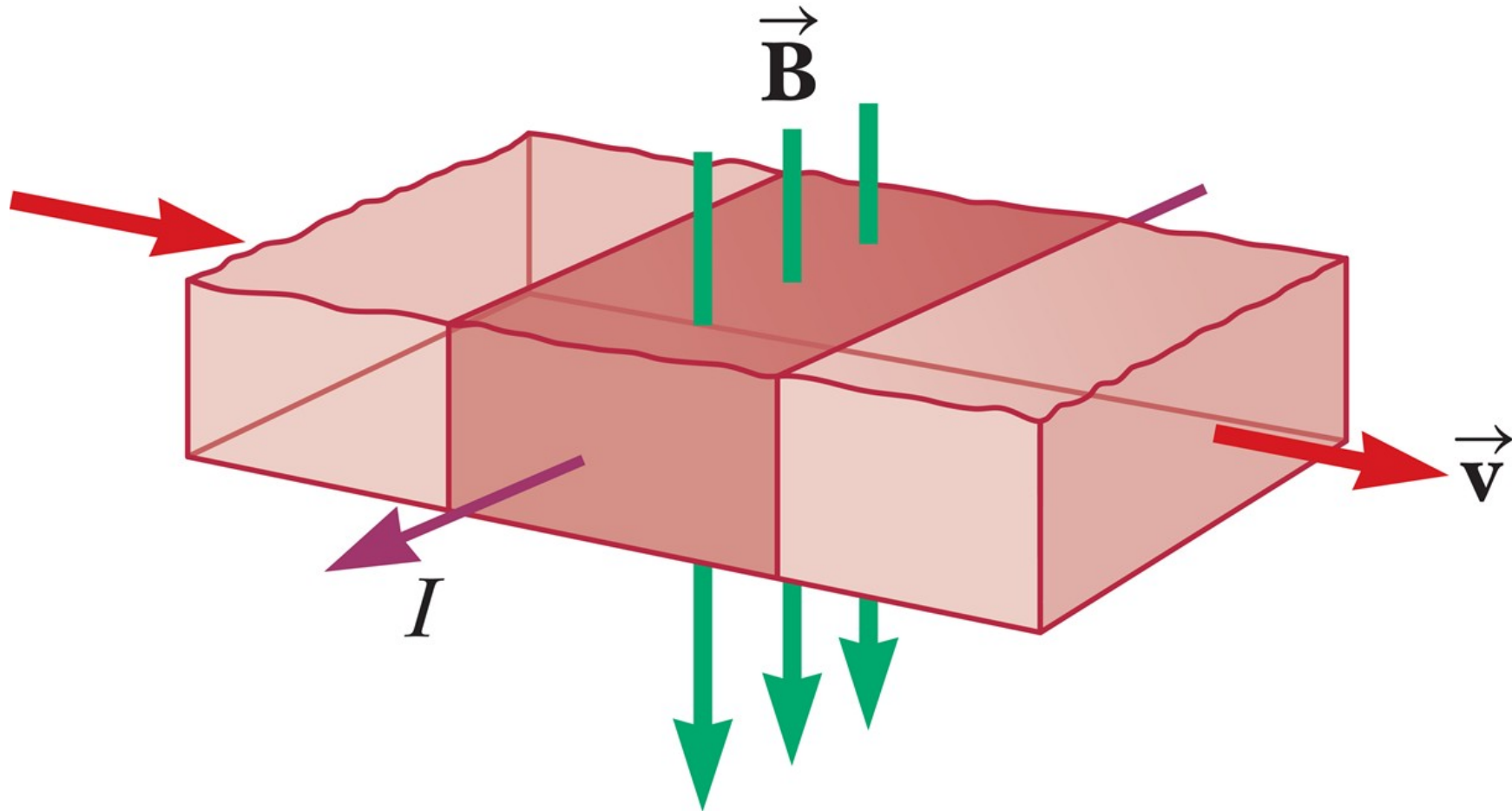


$$F = B I \ell \sin \theta$$

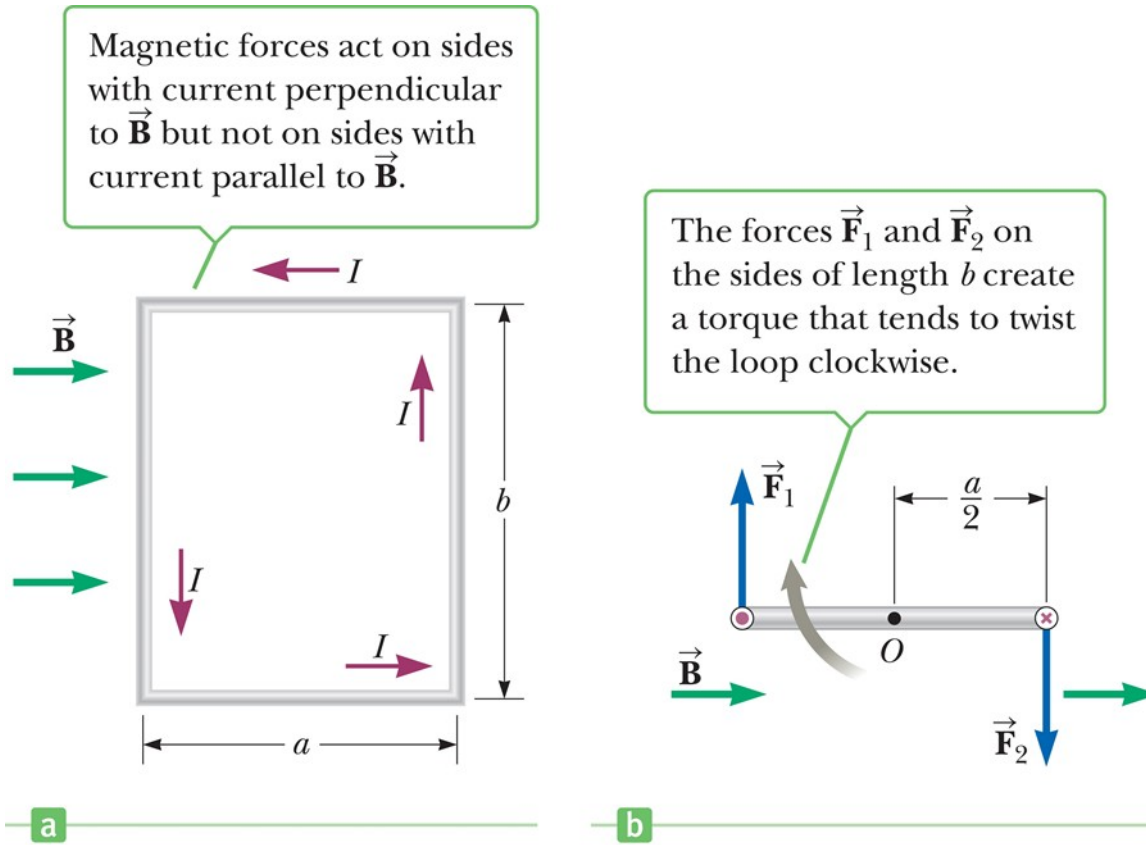
Magnetic Force on a Current-Carrying Conductor (5 of 6)



Magnetic Force on a Current-Carrying Conductor (6 of 6)



Magnetic Torque (1 of 2)

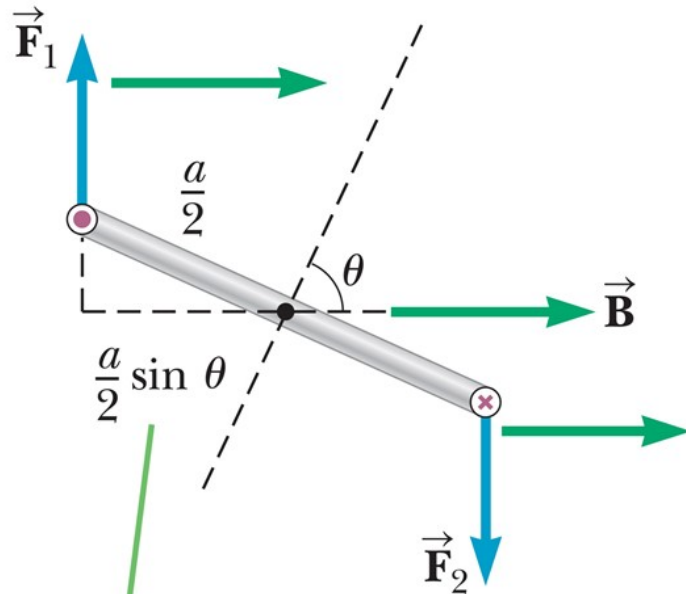


$$F_1 = F_2 = B I b$$

$$\begin{aligned}\tau_{\max} &= F_1 \frac{a}{2} + F_2 \frac{a}{2} \\ &= (B I b) \frac{a}{2} + (B I b) \frac{a}{2} \\ &= B I a b\end{aligned}$$

$$a b = A \quad \rightarrow \quad \tau_{\max} = B I A$$

Magnetic Torque (2 of 2)



If $\vec{B}$ is at an angle θ with a line perpendicular to the plane of the loop, the magnitude of the torque is given by $BIA \sin \theta$.

moment arm: $\left(\frac{a}{2}\right) \sin \theta$

$$\tau = BIA \sin \theta$$

$$\tau = BIAN \sin \theta$$

$$\text{let } \mu = IAN \rightarrow \tau = \mu B \sin \theta$$

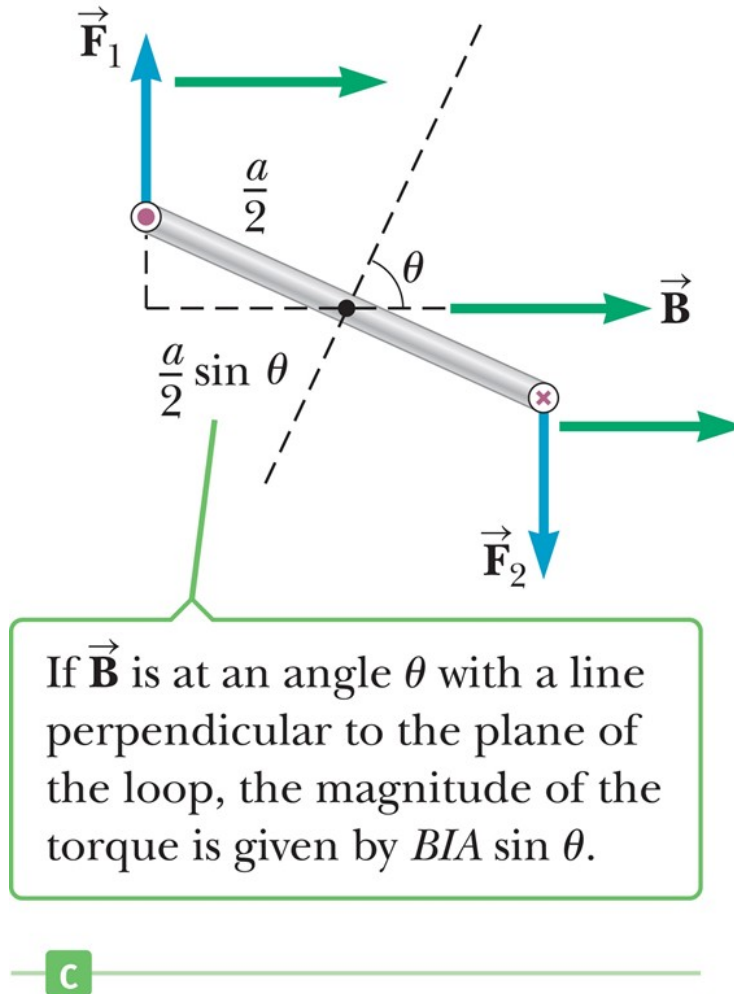
C

Think – Pair – Share 4

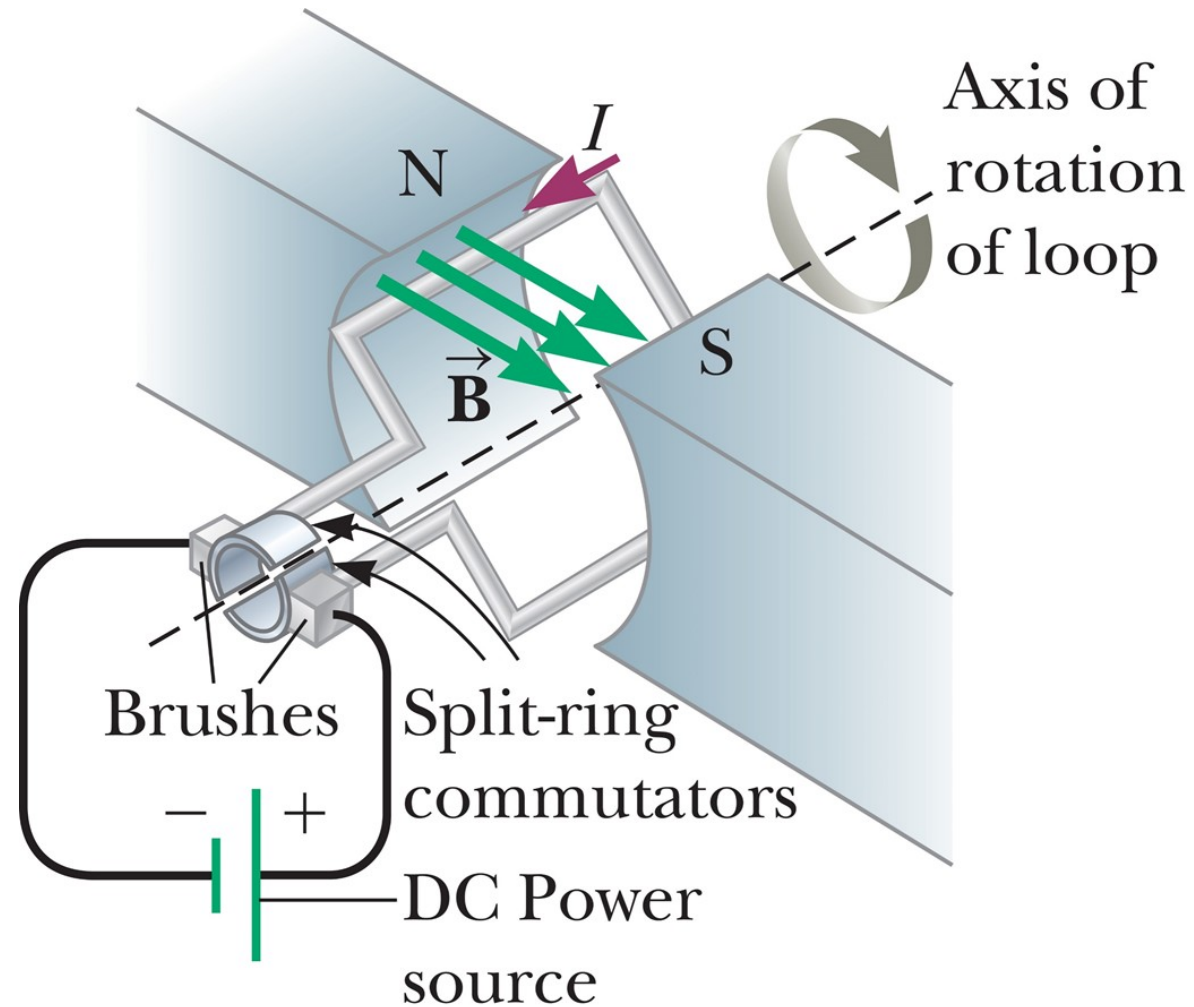
A square and a circular loop with the same area lie in the xy -plane, where there is a uniform magnetic field pointing at some angle θ with respect to the positive z -direction. Each loop carries the same current, in the same direction. Which magnetic torque is larger?

1. the torque on the square loop
2. the torque on the circular loop
3. the torques are the same
4. more information is needed

Electric Motors (1 of 3)



Electric Motors (2 of 3)



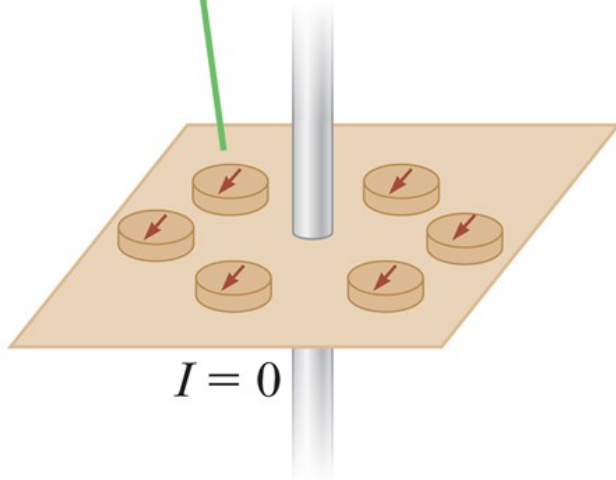
Electric Motors (3 of 3)



John W. Jewett, Jr.

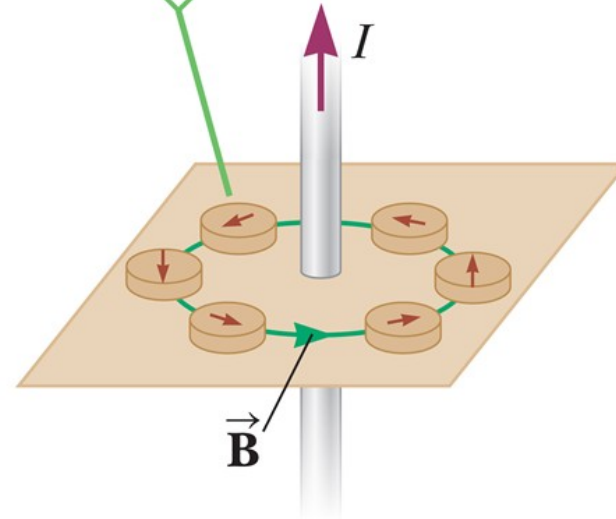
Ampère's Law (1 of 2)

When no current is present in the vertical wire, all compass needles point in the same direction (that of Earth's field).



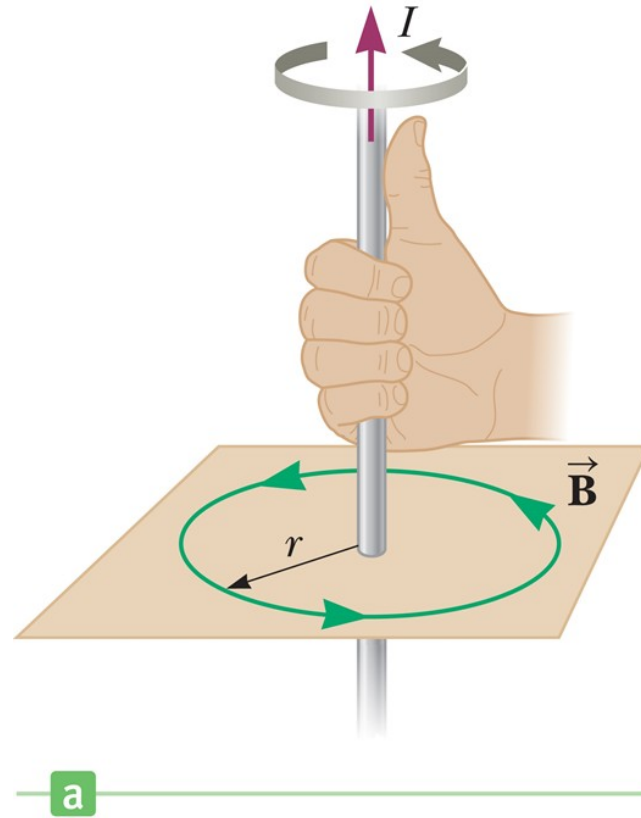
a

When the wire carries a strong current, the compass needles deflect in directions tangent to the circle, pointing in the direction of $\vec{B}$, due to the current.



b

Ampère's Law (2 of 2)



© Richard Megna, Fundamental Photographs

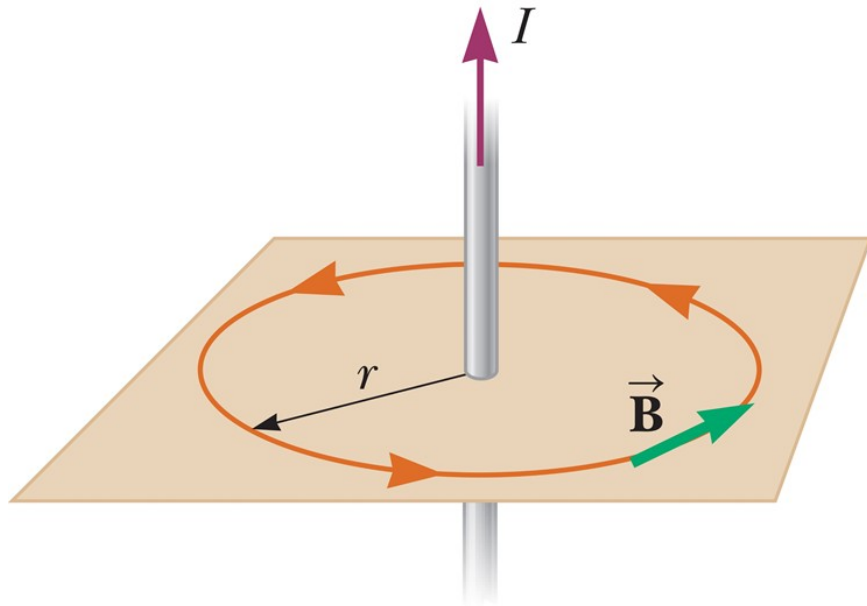
$$B = \frac{\mu_0 I}{2 \pi r} \quad \mu_0 = 4 \pi \times 10^{-7} \text{ T} \cdot \text{m/A}$$

Ampère's Law and a Long, Straight Wire

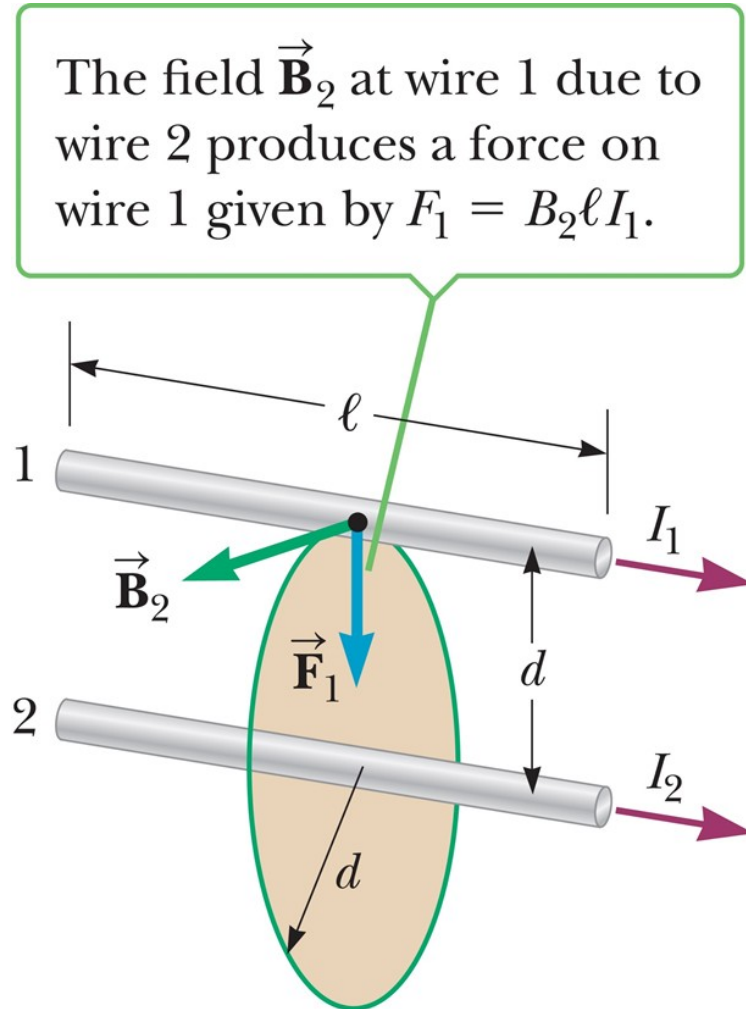
$$\sum B_{||} \Delta \ell = \mu_0 I$$

$$\sum B_{||} \Delta \ell = B_{||} \sum \Delta \ell = B(2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$



Magnetic Force Between Two Parallel Conductors (1 of 2)



$$B_2 = \frac{\mu_0 I_2}{2 \pi d}$$

$$F_1 = B_2 I_1 \ell$$

$$= \left(\frac{\mu_0 I_2}{2 \pi d} \right) I_1 \ell = \frac{\mu_0 I_1 I_2 \ell}{2 \pi d}$$

$$\frac{F_1}{\ell} = \frac{\mu_0 I_1 I_2}{2 \pi d}$$

Magnetic Force Between Two Parallel Conductors (2 of 2)

Definition of an Ampere:

If two long, parallel wires 1 m apart carry the same current and the magnetic force per unit length on each wire is $2 \times 10^{-7} \text{ N/m}$, the current is defined to be 1 A.

Definition of a Coulomb:

If a conductor carries a steady current of 1 A, the quantity of charge that flows through any cross section in 1 s is 1 C.

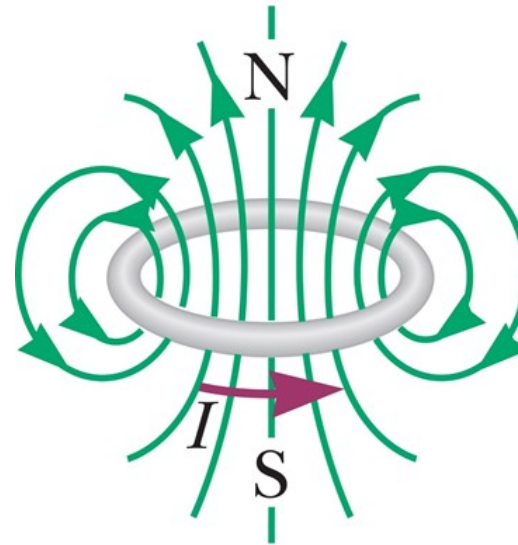
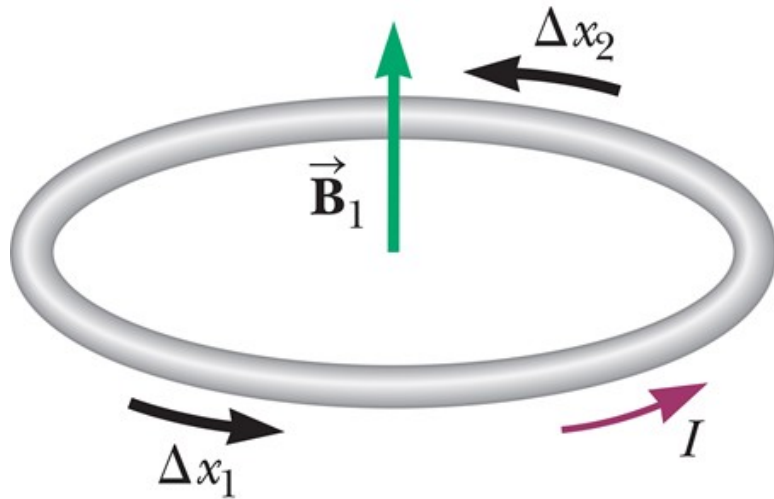
Think – Pair – Share 5

Which of the following actions would double the magnitude of the magnetic force per unit length between two parallel current-carrying wires? Choose all correct answers.

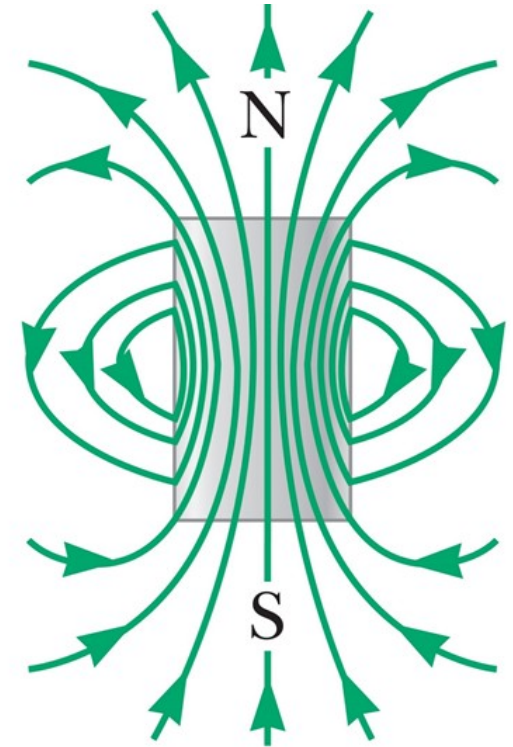
- (a) Double one of the currents.
- (b) Double the distance between them.
- (c) Reduce the distance between them by half.
- (d) Double both currents.

- 1. (a) Only
- 2. (a) and (c)
- 3. (b) and (d)
- 4. (c) and (d)

Magnetic Fields of Current Loops and Solenoids (1 of 2)

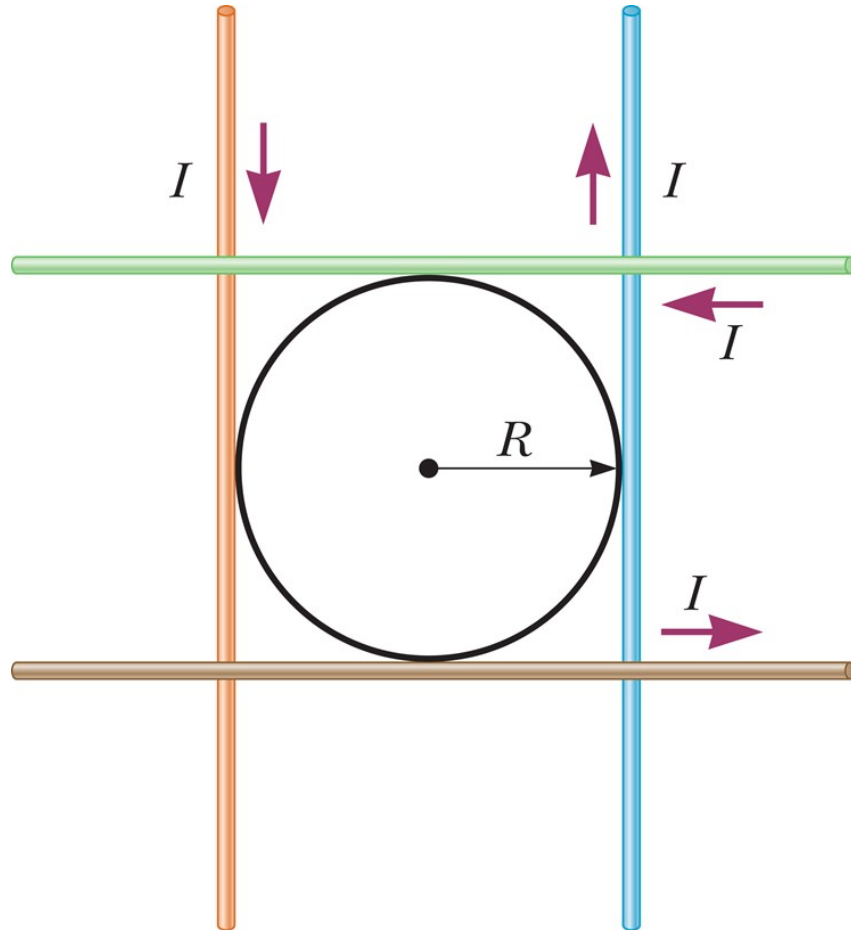


a



b

Magnetic Fields of Current Loops and Solenoids (2 of 2)



$$B = \frac{\mu_0 I}{2R}$$

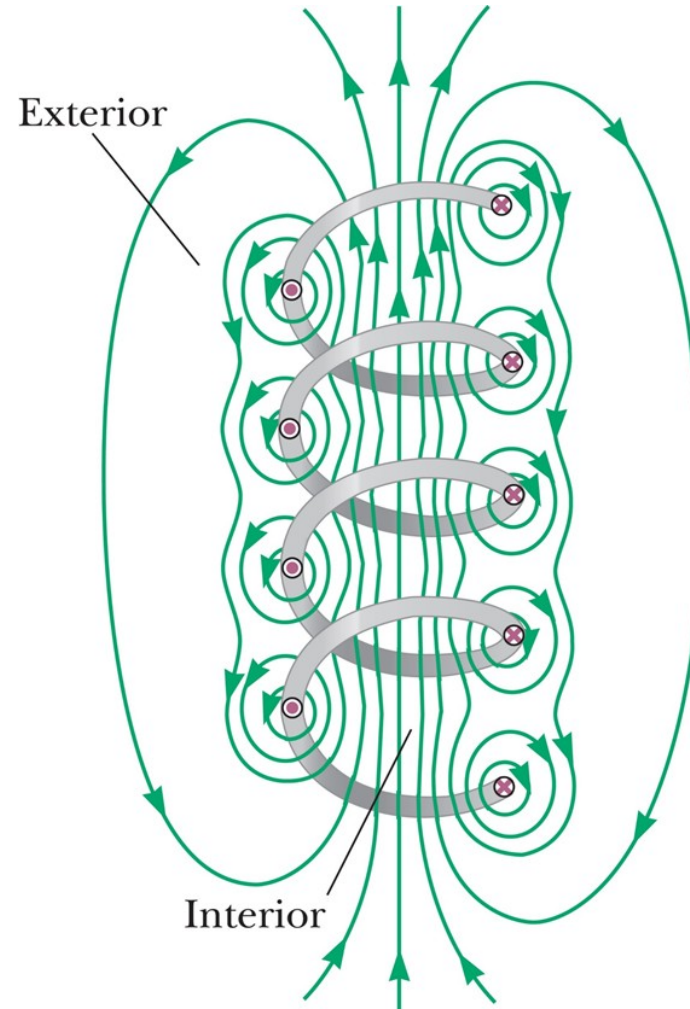
$$B = 4 \times \frac{\mu_0 I}{2\pi R}$$

$$= \frac{4}{\pi} \left(\frac{\mu_0 I}{2R} \right)$$

$$= (1.27) \left(\frac{\mu_0 I}{2R} \right)$$

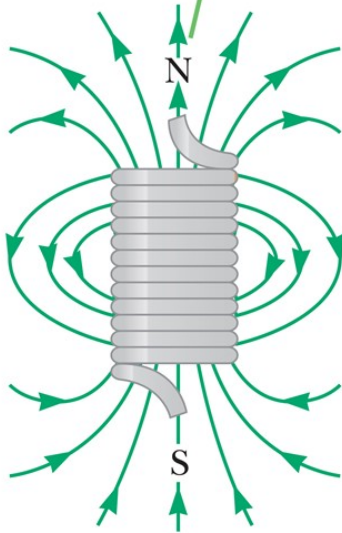
$$B = N \frac{\mu_0 I}{2R}$$

Magnetic Field of a Solenoid (1 of 5)

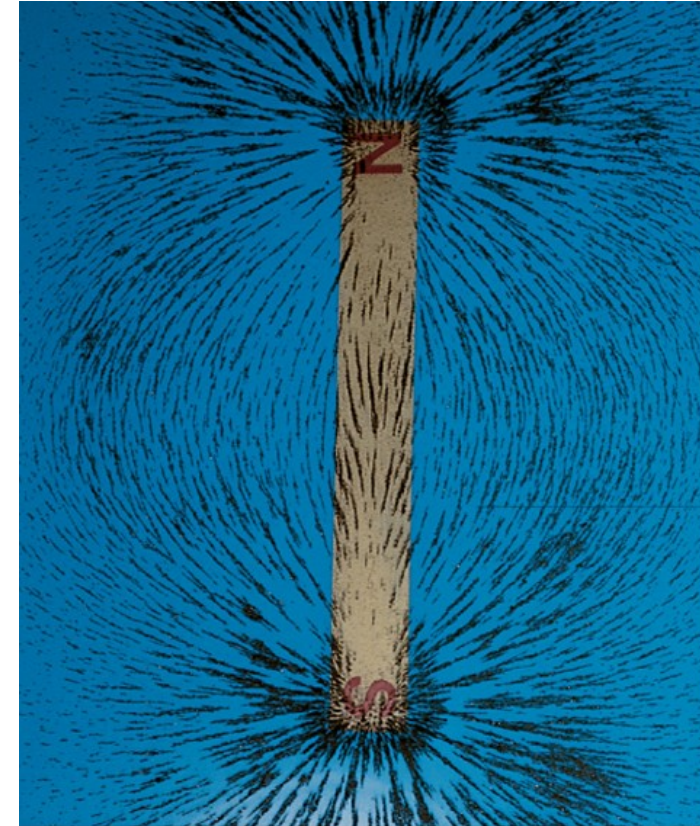


Magnetic Field of a Solenoid (2 of 5)

The magnetic field lines resemble those of a bar magnet, meaning that the solenoid effectively has north and south poles.



a

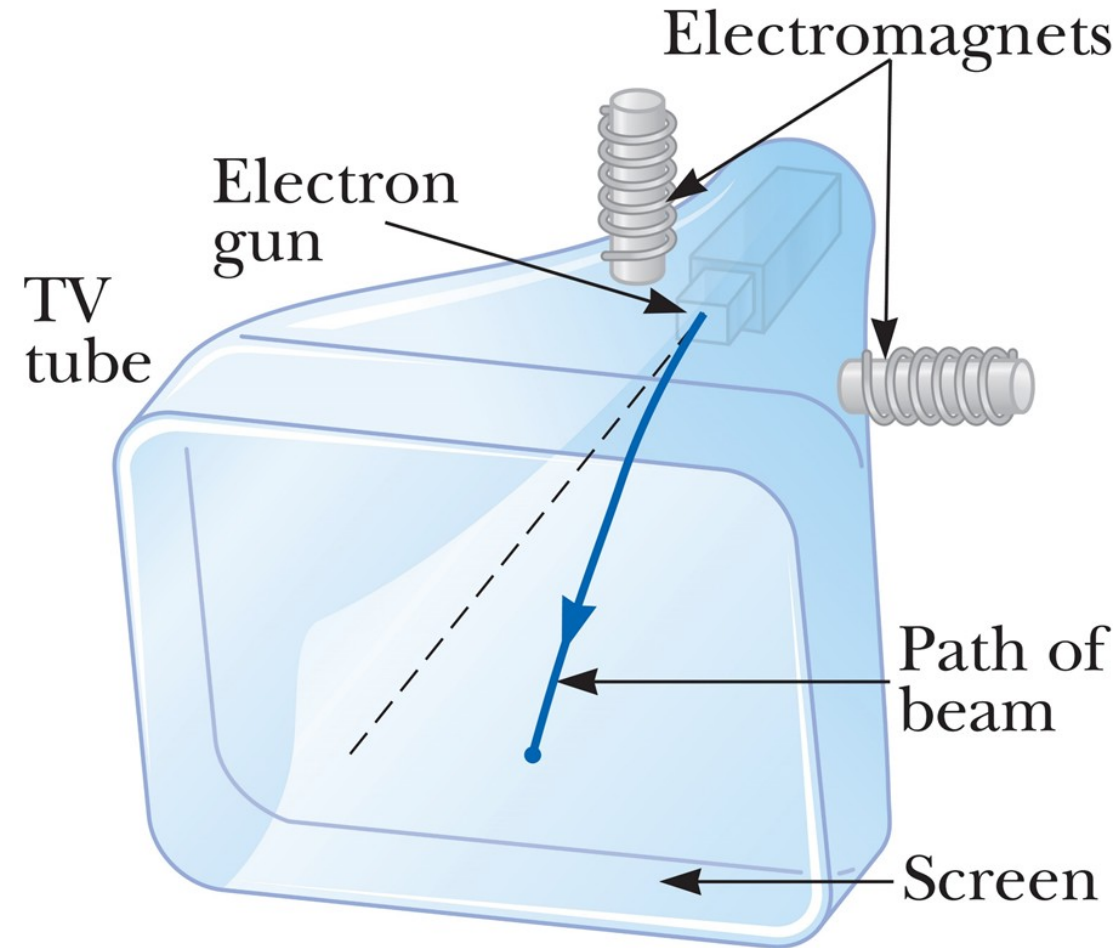


Courtesy of Henry Leap and Jim Lehman

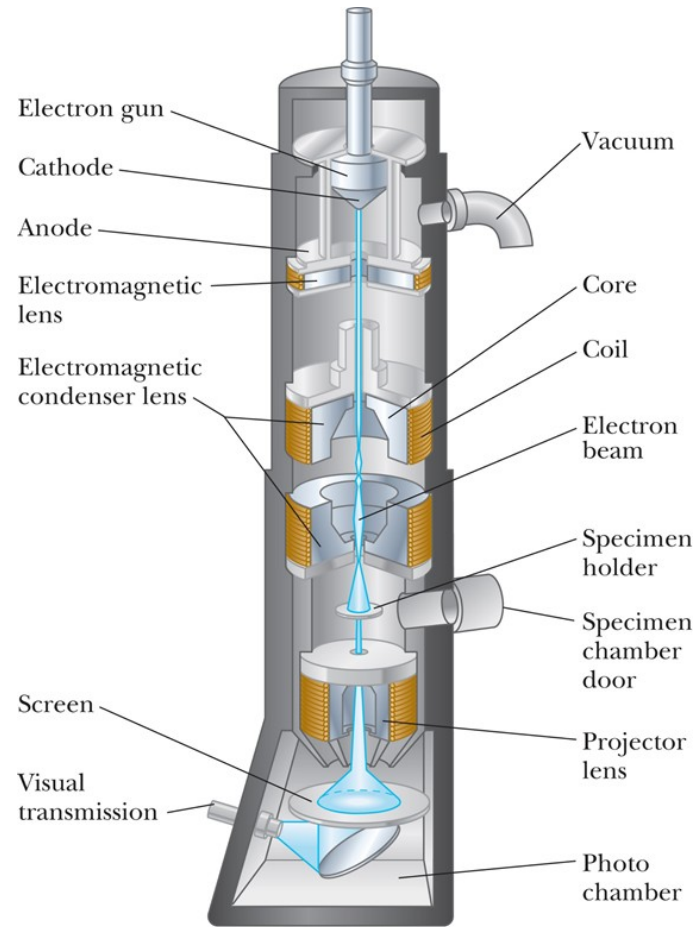
b

$$B = \mu_0 n I$$

Magnetic Field of a Solenoid (3 of 5)



Magnetic Field of a Solenoid (4 of 5)



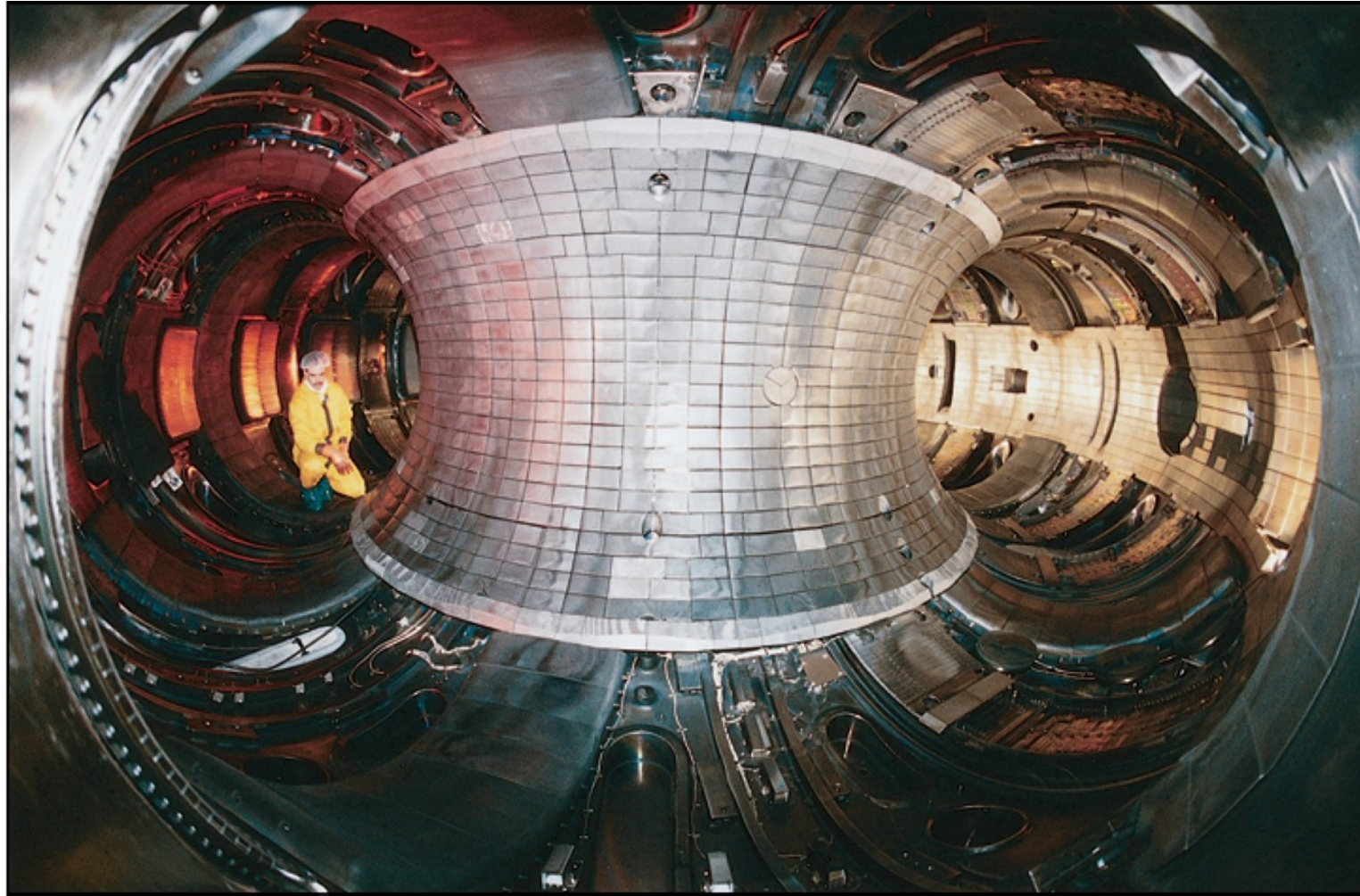
a



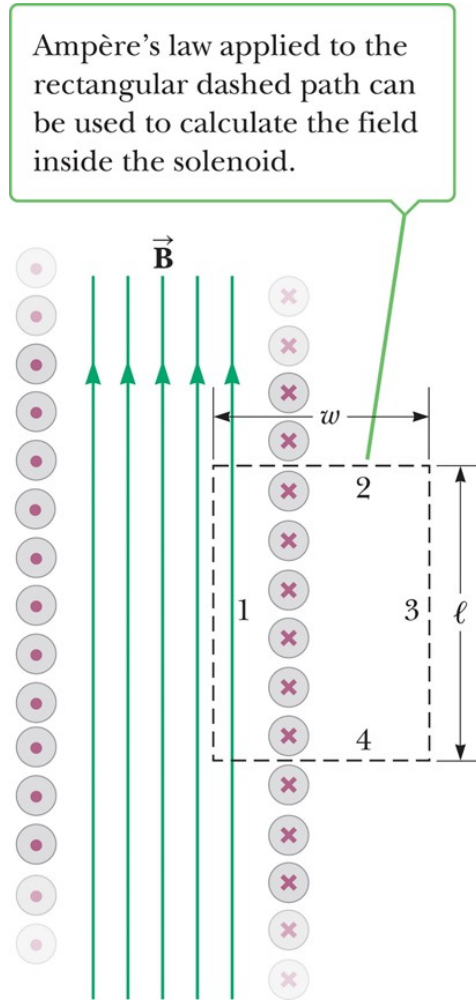
b

Magnetic Field of a Solenoid (5 of 5)

Courtesy of Princeton Plasma Physics Laboratory



Ampère's Law Applied to a Solenoid

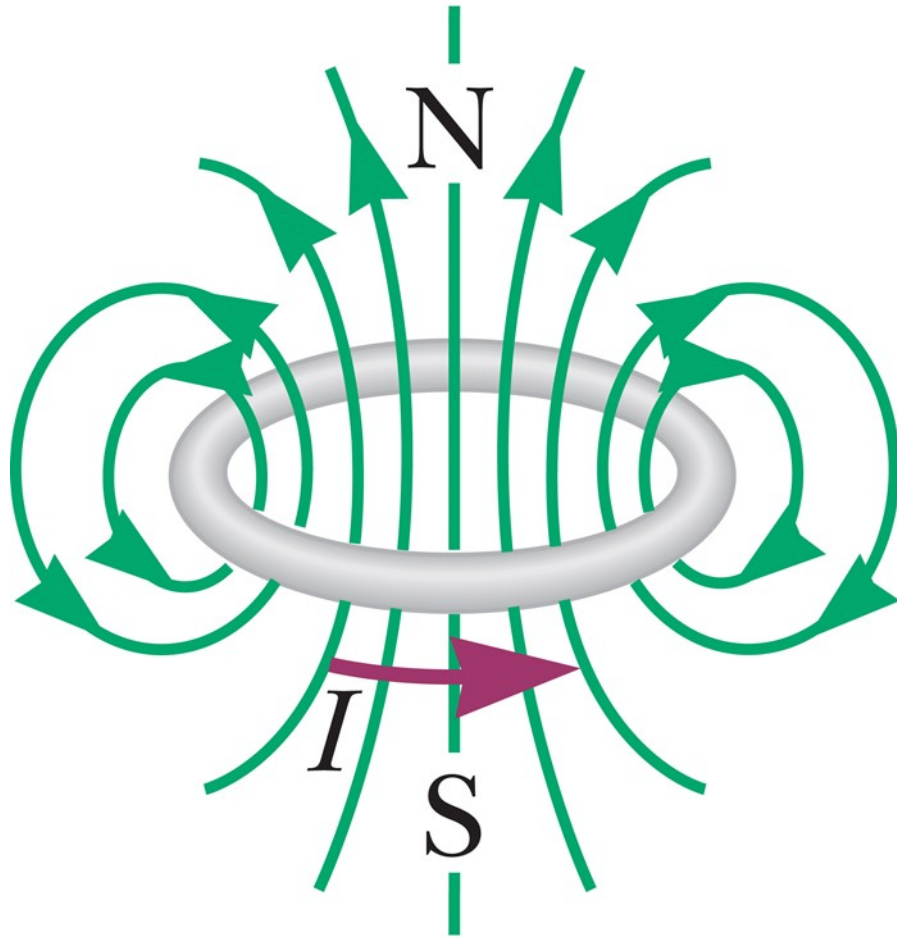


$$\sum B_{\parallel} \Delta \ell = \mu_0 I$$

$$\sum B_{\parallel} \Delta \ell = BL$$

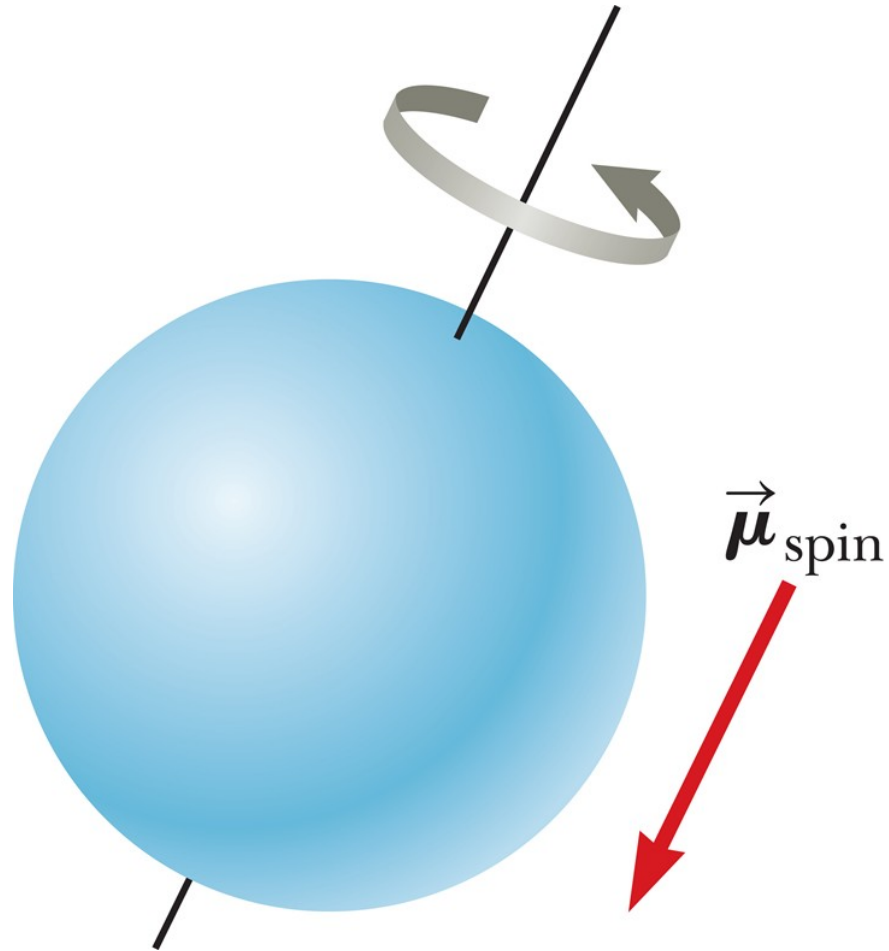
$$B = \mu_0 \frac{N}{L} I = \mu_0 n I$$

Magnetic Domains (1 of 4)



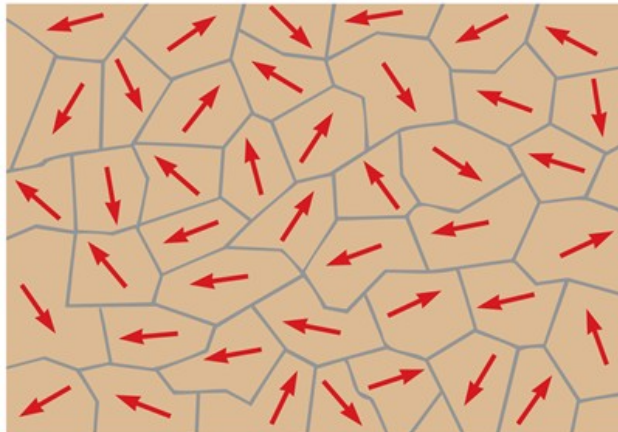
$$\frac{1.6 \times 10^{-19} \text{ C}}{10^{-16} \text{ s}} = 1.6 \times 10^{-3} \text{ A}$$

Magnetic Domains (2 of 4)



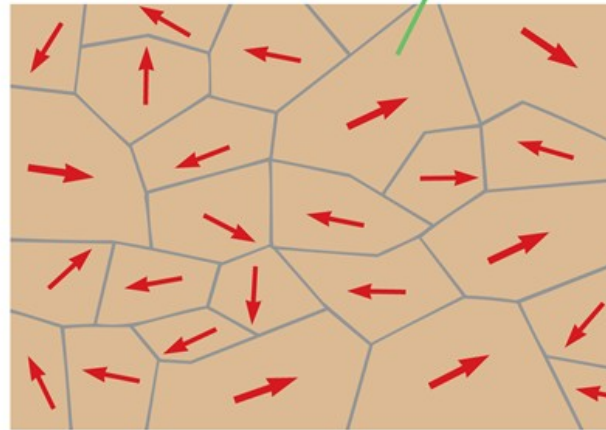
Magnetic Domains (3 of 4)

Random orientation of domains in an unmagnetized substance.



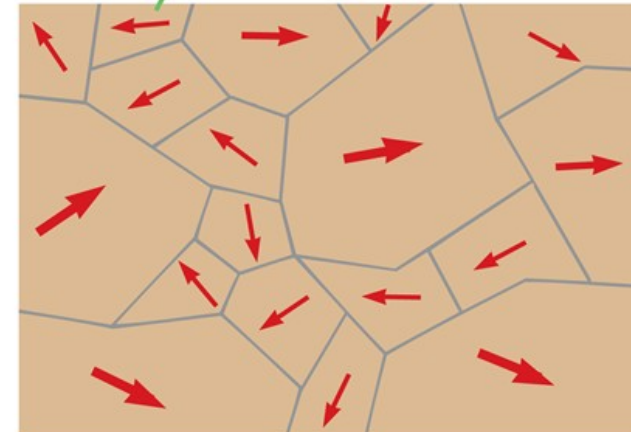
a

When an external magnetic field $\vec{B}$ is applied, the domains tend to align with the magnetic field.



b

As the field strengthens, domains aligned with $\vec{B}$ tend to grow larger while those not aligned grow smaller.



c

Magnetic Domains (4 of 4)



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Types of Magnetic Materials

High Field Magnet Laboratory, University
of Nijmegen, The Netherlands

